EFWWAV1 Module Datasheet

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1 Overview:

The E-Factor IoT module is a multifunctional module designed for versatility and flexibility, supporting various communication protocols like Wi-Fi, RF, and PLC.

Its capability to connect with multiple end devices using the UART protocol and update data to the cloud through the E-Factor protocol or Efactor Mobile application makes it a robust choice for various IoT applications. Ensuring secure, reliable, and efficient operations, the module stands out as a comprehensive solution in the IoT ecosystem.

2 IoT Module Specifications:

1) Efactor IoT Module dimensions 22mm X 17mm.
2) Module Interface connector pitch 1.27mm
3) Efactor IoT module can support multiple baud rates like 4800, 9600, 38400, 57600, 115200, 230400. The default baud rate is 115200.
4) This IoT module is compatible with wireless communications via Wi-Fi.

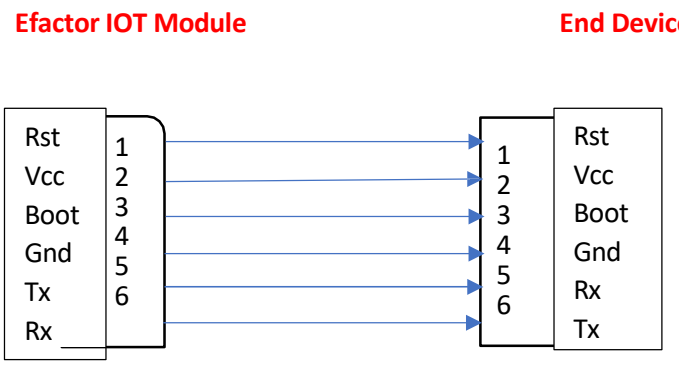
3 Features:

1) Embedded low-power 32-bit CPU, which can also function as an application processor (The clock rate supports 80 MHz and 160 MHz)
2) Working voltage: 3.0 to 3.6 V
3) Peripherals: 1 universal asynchronous receiver/transmitter (UART)
4) Wi-Fi connectivity
   - IEEE 802.11 b/g/n
   - Channels 1 to 14@2.4 GHz
   - Support WEP/WPA/WPA2/WPA2 PSK (AES) security mode
   - Up to +20 dBm output power in 802.11b mode
   - Support STA/AP/STA+AP working mode
   - Support SmartConfig and AP network configuration manners for Android
   - Onboard PCB antenna with a gain of 3.0 dBi
   - Working temperature: -20 to +85℃

4 Efactor IoT Module Interface Pin Description:

S.No	Pin	Description
1	Reset	Optional
2	VCC	3.3V ,0.2-0.5A(max)
3	Boot	Optional
4	Gnd	common ground
5	Tx	UART communication Transmitter Pin
6	Rx	UART communication Receiver Pin

5 EFACTOR IoT Module to End Device Interface:







6 Efactor IOT Module:

1) Connector pin description is described in above sections.
2) By scanning the QR code located on the top of the module, users can add that module or specific device to the APP.

7 Module Function Details:

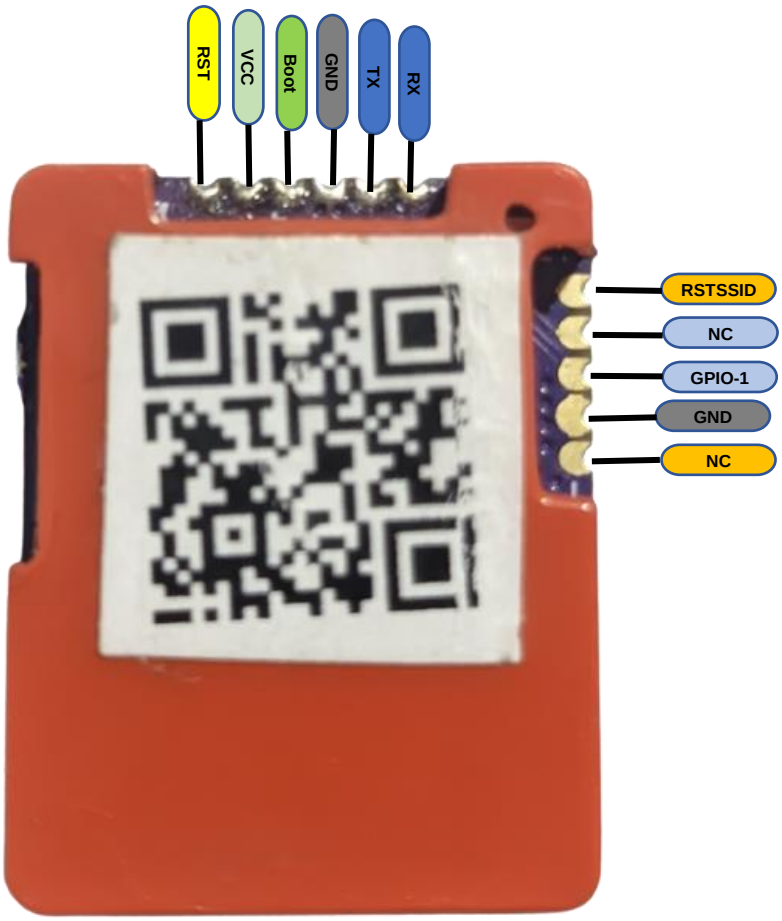
1) Module is connected to end device with UART. It is sending and receiving Messages from EFACTOR Gateway by using EFACTOR protocol and also connected to mobile APP.
2) This module can update firmware with OTA from EFACTOR Cloud.
3) The UART frame size of one command or data should be maximum 80 bytes, If the data is more than 80 bytes, then it follows the below format

Example:

CMD	No.of Frames	Frame Index	Data (<=80 bytes)
DE	04	00	Data

4) The gap of one frame to another frame UART receiving and UART transmitting Should maintain 300ms.

8 Efactor IoT Module Pin-Out:



9 Reset SSID and Password pin:

1) Grounding the RSTSSID Pin:

To initiate the SSID and password reset, ground the RSTSSID pin of the module.

Maintain the grounding connection for a duration of a 3 seconds.

2) Reset Confirmation:

The module will acknowledge the reset operation through predefined indicators, signalling the successful initiation of the reset process.

